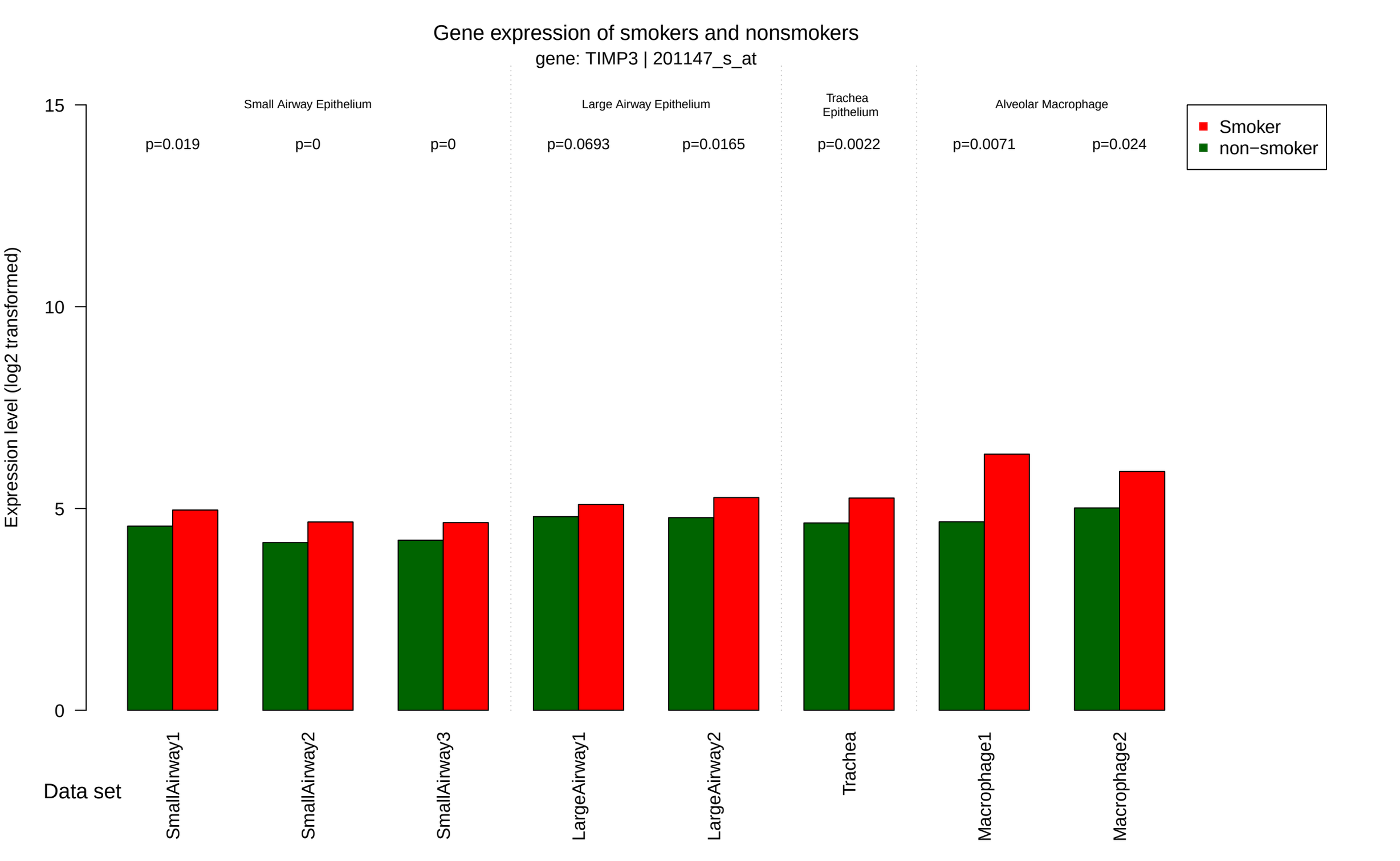
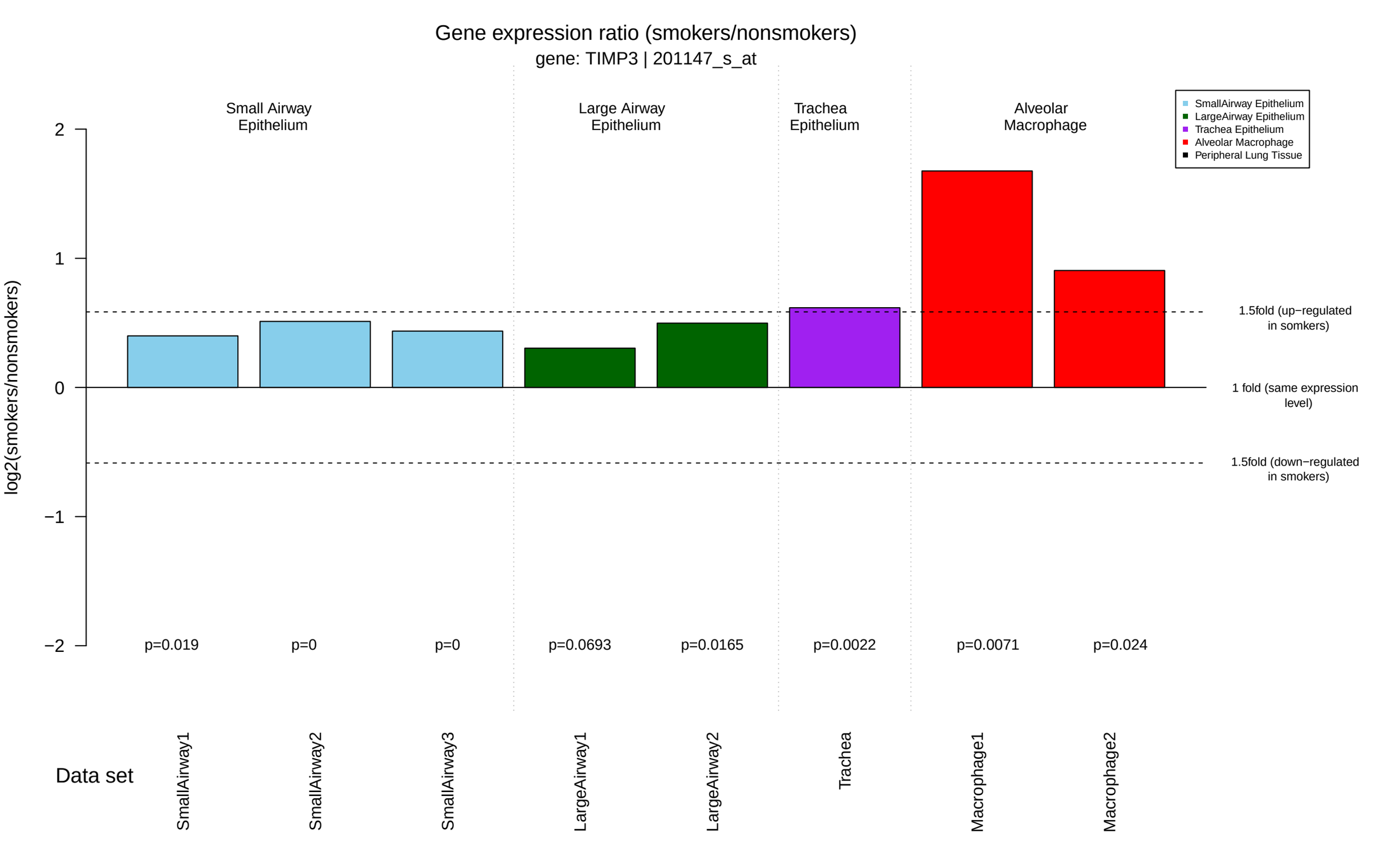
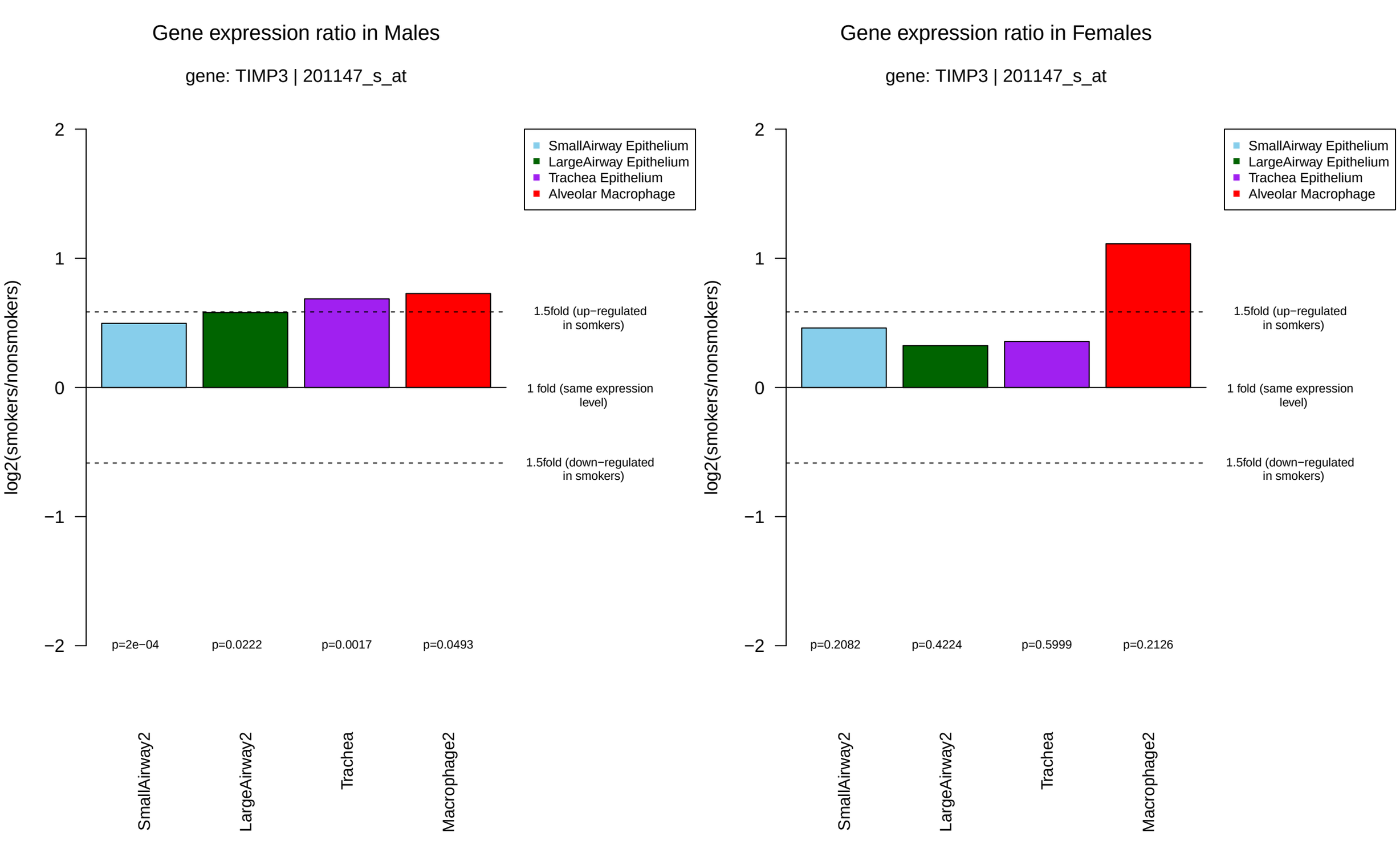
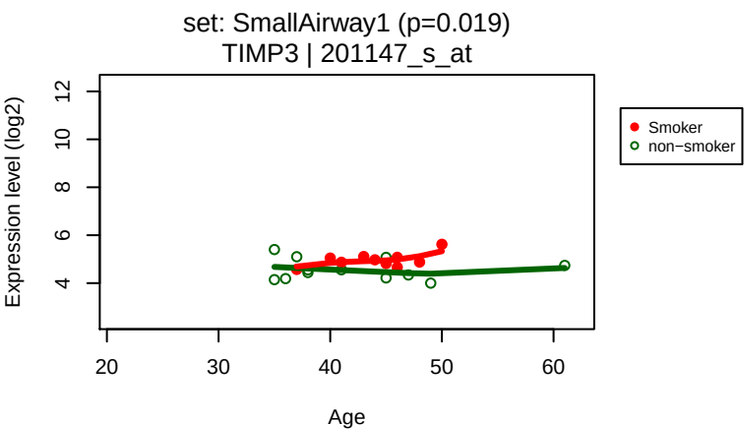
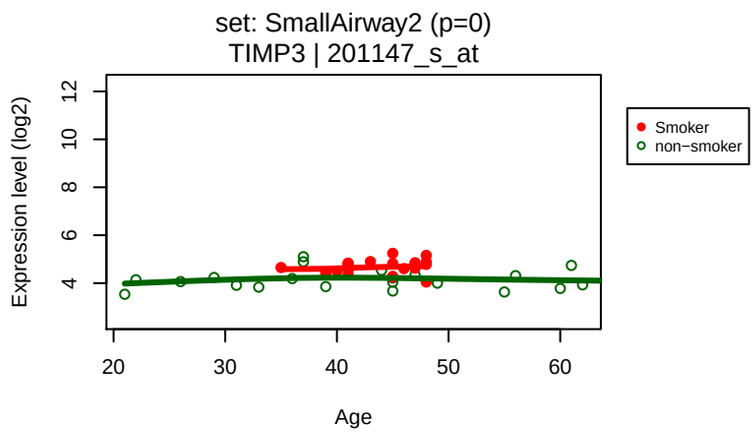
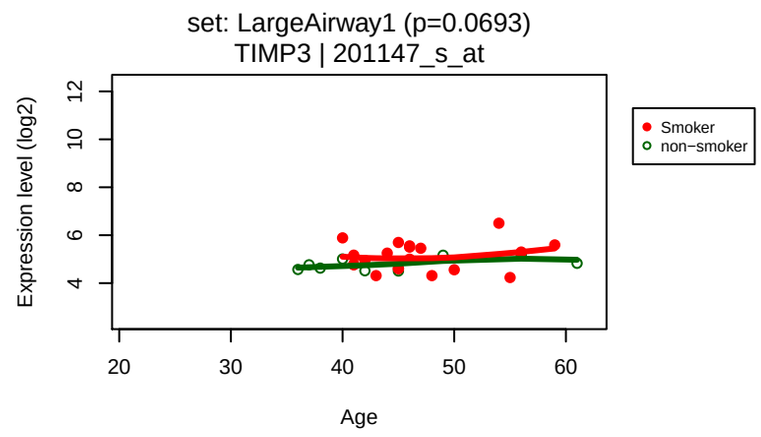

Gene expression of smokers and nonsmokers
gene: TIMP3 | 201147_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.019
p=0
p=0
p=0.0693
p=0.0165
p=0.0022
p=0.0071
p=0.024
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

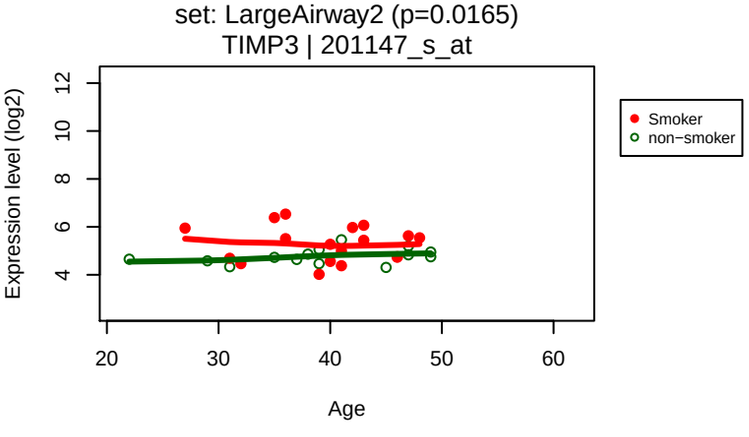

Gene expression ratio (smokers/nonsmokers)
gene: TIMP3 | 201147_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.019
p=0
p=0
p=0.0693
p=0.0165
p=0.0022
p=0.0071
p=0.024
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

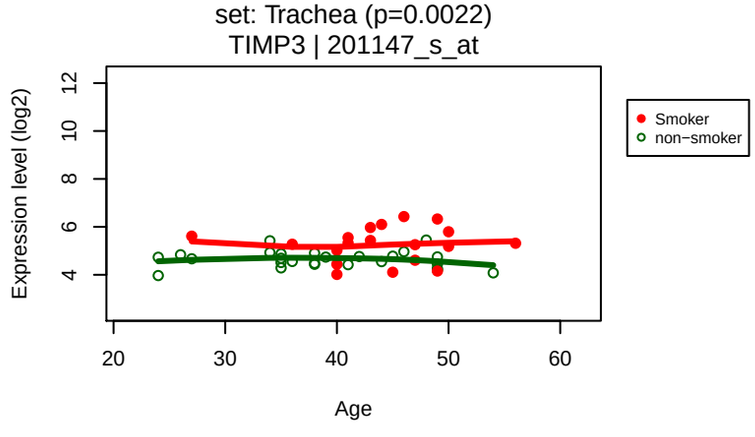

Gene expression ratio in Males
gene: TIMP3 | 201147_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=2e−04
p=0.0222
p=0.0017
p=0.0493
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TIMP3 | 201147_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2082
p=0.4224
p=0.5999
p=0.2126
SmallAirway2
LargeAirway2
Trachea
Macrophage2

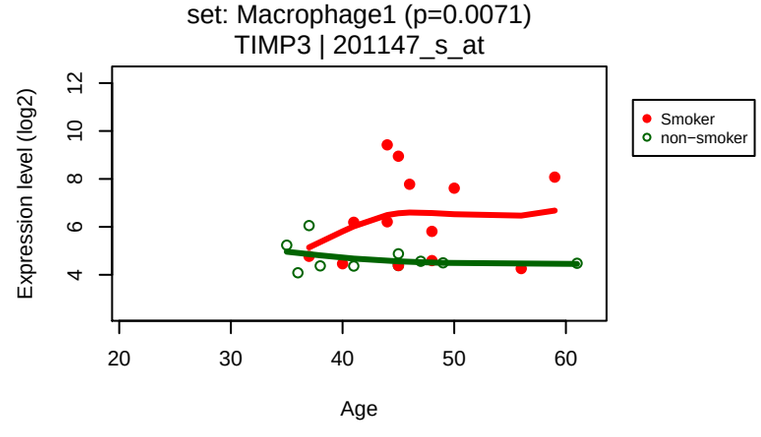

set: SmallAirway1 (p=0.019)
TIMP3 | 201147_s_at
Expression level (log2)
Age
Smoker
non−smoker

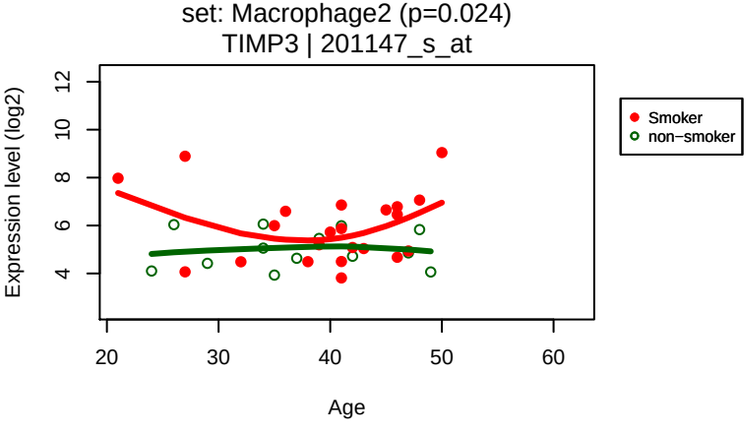

set: SmallAirway2 (p=0)
TIMP3 | 201147_s_at
Expression level (log2)
Age
Smoker
non−smoker

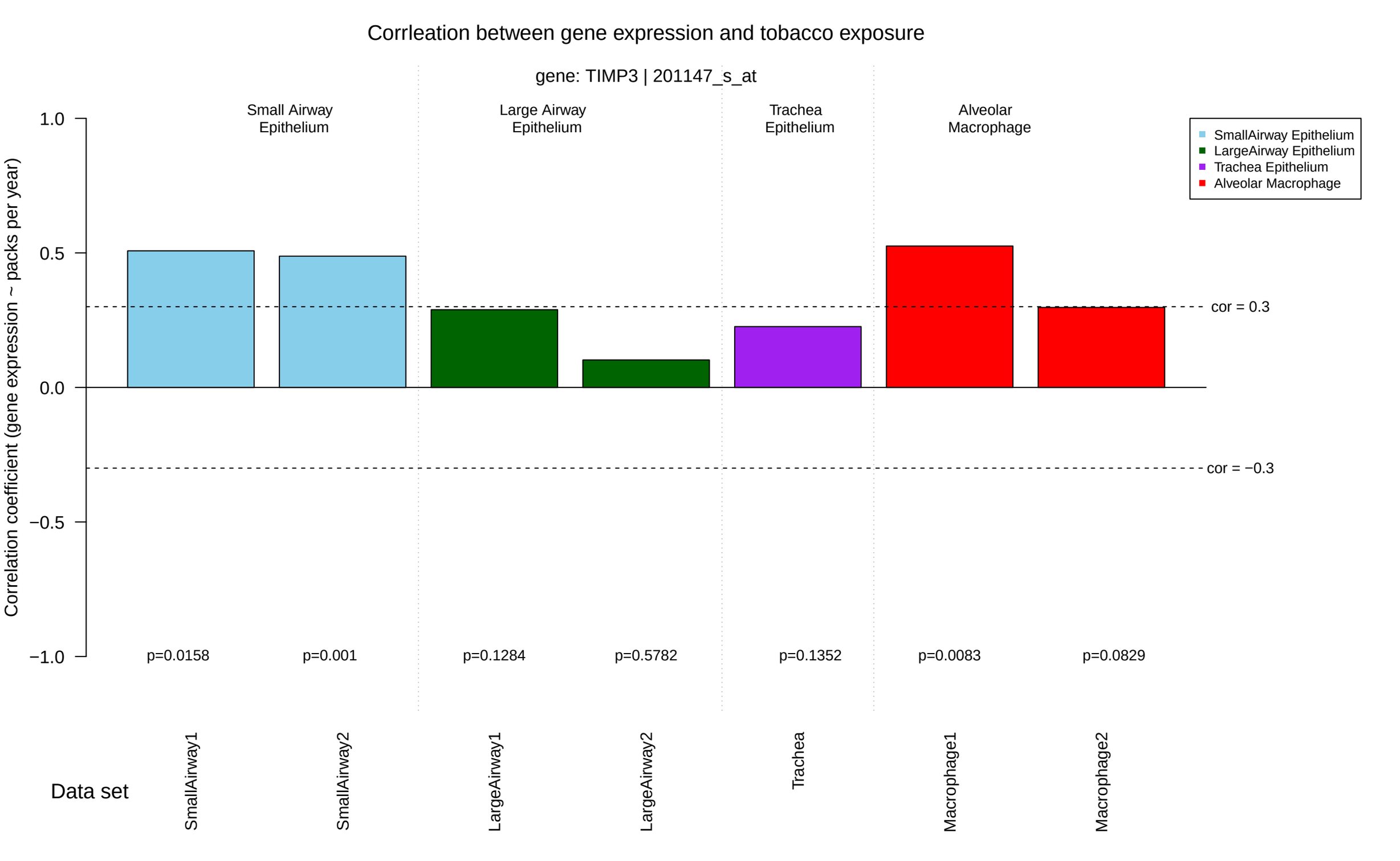

set: LargeAirway1 (p=0.0693)
TIMP3 | 201147_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0165)
TIMP3 | 201147_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0022)
TIMP3 | 201147_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0071)
TIMP3 | 201147_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.024)
TIMP3 | 201147_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TIMP3 | 201147_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0158
p=0.001
p=0.1284
p=0.5782
p=0.1352
p=0.0083
p=0.0829
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2